Chapter 5

Zig-Zag Product

5.1 Definition of the Zig-Zag Product

Definition 5.1. Graph X, e an edge in X, v a vertex incident to e. Define $e()$ as:

- $e(v) = v$ if e is a loop;
- $e(v) = w$, where w is the other vertex, if e is not a loop

That is, $e()$ takes the vertex adjacent to v through edge e.

Definition 5.2. Zig-Zag product (graph)

- Let Y be d_Y-regular, X be $|Y|$-regular
- Vertex sets V_X, V_Y; edge multisets E_X, E_Y, multiple edges treated as distinct elements
- For each vertex $v \in V_X$, let $E_v = \{e \in E_X \,|\, v \text{ is an endpoint of } e\}$, let $L_v : V_Y \to E_v$ be a bijection (which always exists since X is $|Y|$-regular, making $|V_Y| = |E_v|$)
- Call L_v the labeling at v. Let L be the set $\{L_v \,|\, v \in V_Y\}$, call L the labeling from Y to X.
- Define the zig-zag product $X \textcircled{z} Y$ with labeling L as follows
  - Vertex set: $X \times Y$
  - Multiplicity (number) of edges between (x_1, y_1) and (x_2, y_2): number of ordered pairs $(z_1, z_2) \in E_Y \times E_Y$ s.t.
    1. y_1 is an endpoint of z_1, y_2 is an endpoint of z_2
    2. $L_{x_1}(z_1(y_1)) = L_{x_2}(z_2(y_2))$

Lemma 5.3. A comprehensible way of finding edges in zig-zag product (graph): Let (x_1, y_1) be a vertex in $X \textcircled{z} Y$, now we find one of its adjacent vertex and the corresponding edge. Note that an adjacent vertex of (x_1, y_1) can correspond to multiple edges, which are treated as distinct elements.

Step 0 (x_1, y_1)

Step 1 (Zig)
- Choose edge z_1 in Y that is incident to y_1
- Get $(x_1, z_1(y_1))$

Step 2 (-)
- Let $e = L_{x_1}(z_1(y_1))$, note $L_{x_1} : V_Y \to E_{x_1}$. Note x_1 must be an endpoint of e
- Let x_2 be the other endpoint of edge e
- Let $y' = L_{x_2}^{-1}(e)$
- Get (x_2, y')

Step 3 (Zag)
- Choose z_2 in Y that is incident to y'
- Let $y_2 = z_2(y')$
- Get (x_2, y_2)

Remark
Accounting practice:

- d_Y choices of z_1 (edges incident to y_1)
- $z_1(y_1)$ is unique (only two vertices for an edge)
- $e = L_{x_1}(z_1(y_1))$ is unique (bijection)
- x_2 is the endpoint of e other than x_1, unique
- $y' = L_{x_2}^{-1}(e)$ is unique (bijection)
- d_Y choices of z_2 (edges incident to y')
- $y_2 = z_2(y')$ is unique (only two vertices for an edge)

Hence, the zig-zag product is d_Y^2-regular

Definition 5.4. Let v be a vertex in X, define the v-cloud in $X \textcircled{z} Y$ to be $\{(v, y) \mid y \in V_Y\}$

Remark
It can be thought that we get $X \textcircled{z} Y$ by replacing each vertex v of X with the v-cloud.

Example
In the following example, we denote the labeling L from Y to X by labeling edges in X with vertices of Y. Therefore, in X, each vertex has its incident edges labeled.

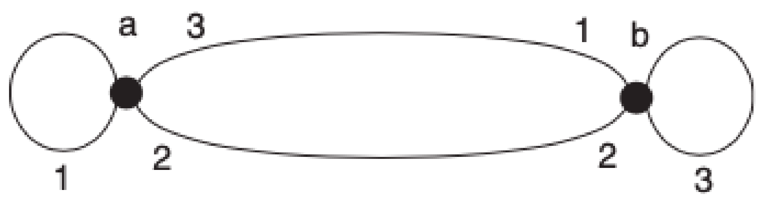

Figure 5.4 X

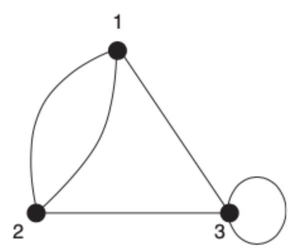

Figure 5.5 Y

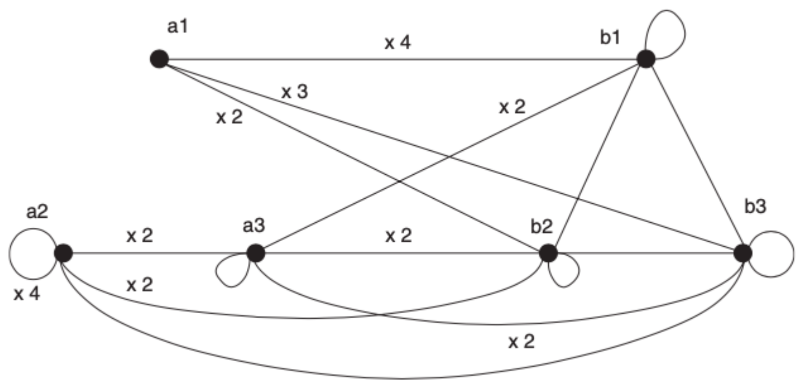

Figure 5.6 $X{\text{Ⓩ}}_M Y$

Remark
Zig-Zag products depend on the choice of labeling: different labeling could generate different graphs.

5.2 Adjacency Matrices and Zig-Zag Product

In this section, we express the adjacency operator of $X \textcircled{z} Y$ in terms of the adjacency operators of X and Y.

Definition 5.5. Zig&Zag graph (Z) and Hyphen graph (H)

- Graph Z
- Vertex Set: $V_X \times V_Y$
- Multiplicity between (x_1, y_1) and (x_2, y_2):
  - multiplicity of edge between y_1 and y_2, if $x_1 = x_2$;
  - 0, if $x_1 \neq x_2$

- Graph H
- Vertex Set: $V_X \times V_Y$
- Multiplicity between (x_1, y_1) and (x_2, y_2):
  - 1, if $L_{x_1}(y_1) = L_{x_2}(y_2)$;
  - 0, otherwise

Remark

- Z is d_Y-regular: fix (x_1, y_1), total multiplicity of incident edges is d_Y
- H is 1-regular (see Prop 5.6)

Proposition 5.6. $X \textcircled{z} Y = Z \cdot H \cdot Z$

Proof

- Same vertex set, suffice to track edges. Start with (x_1, y_1), track its incident edge e_0. On the RHS, we go through $e_1 \in Z, e_2 \in H, e_3 \in Z$, and $\text{mul}(e_0) = \text{mul}(e_1)\text{mul}(e_2)\text{mul}(e_3)$.
- Zig-Step (Z): Choose z_1, e_1 is between (x_1, y_1) and $(x_1, z_1(y_1))$, and $\text{mul}(e_1) = d_Y$ in total.

- Hyphen-Step (H): No choice, everything fixed. e_2 is between $(x_1, z_1(y_1))$ and (x_2, y'). Since $L_{x_1}(z_1(y_1)) = e$, $L_{x_2}(y') = L_{x_2}(L_{x_2}^{-1}(e)) = e$, $\mathrm{mul}(e_2) = 1$ in total.

- Zag-Step (Z): Choose z_2, e_2 is between $(x_2, y'))$ and $(x_2, z_2(y'))$, and $\mathrm{mul}(e_3) = d_Y$ in total.

- By go through $e_1 \in Z, e_2 \in H, e_3 \in Z$ we indeed get e_0 in $X \textcircled{Z} Y$ as an edge incident to (x_1, y_1). Also, degree of (x_1, y_1) calculated with this approach is same as d_Y^2 in $X \textcircled{Z} Y$, done.

Example
This example shows how Z and H look like.

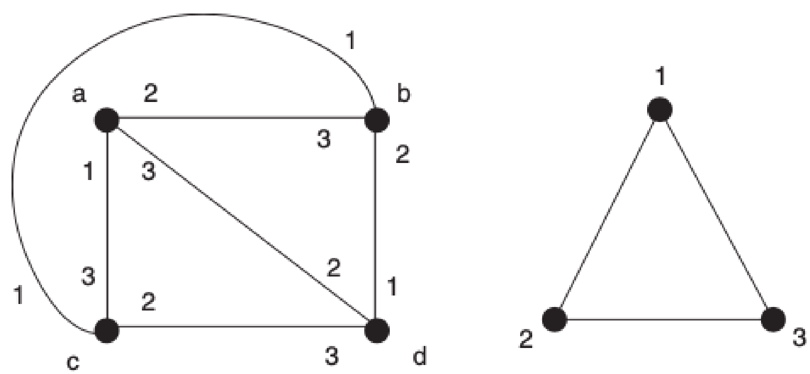

Figure 5.7 A graph X (left) and a graph Y (right), with a labeling from Y to X

Definition 5.7.

- Fix an ordering $x_1, \ldots, x_n$ of the vertices of X and an ordering $y_1, \ldots, y_{d_X}$ of the vertices of Y. Order the vertices of $X \textcircled{Z} Y$ lexicographically, i.e. $x_1y_1, \ldots, x_1y_{d_X}, x_2y_1, \ldots, x_2y_{d_X}, \ldots, x_ny_1, \ldots, x_ny_{d_X}$

- Let $\tilde{B}, \tilde{A}, \tilde{M}$ be the adjacency matrices of Z, H, and $X \textcircled{Z} Y$, in terms of this ordering.

- Let A, B be the adjacency matrices of X, Y, in terms of this ordering.

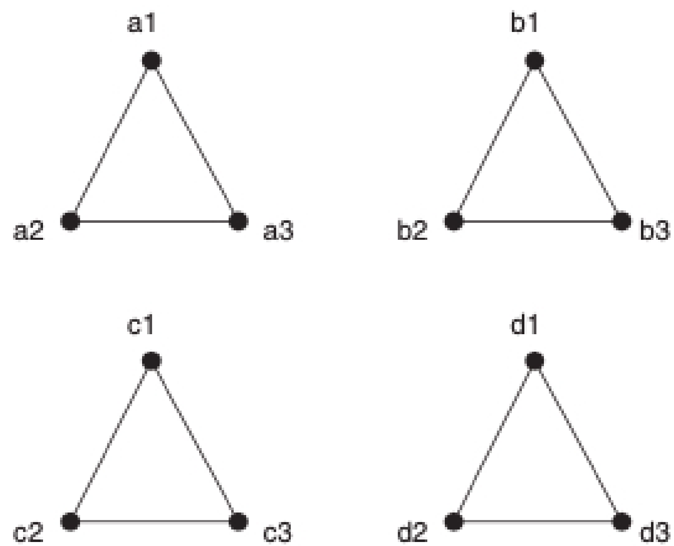

Figure 5.8 The "zig" graph Z

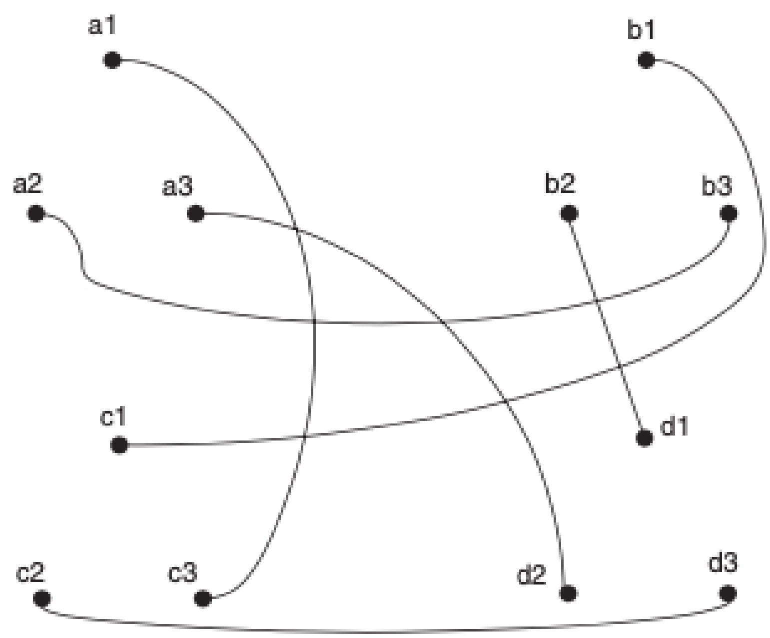

Figure 5.9 The "hyphen" graph H

Corollary 5.8. $\tilde{M} = \tilde{B}\tilde{A}\tilde{B}$

Proof
Since all matrices have same ordering of the same vertex set, Prop 1.32

Proposition 5.9. $\tilde{B}$ equals the $|X| \cdot |Y| \times |X| \cdot |Y|$ block diagonal matrix

$$\begin{pmatrix} B & 0 & \dots & 0 \\ 0 & B & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & B \end{pmatrix}$$

Proof
Each block is a $|X| \times |X|$ matrix, and there are $d_X \times d_X$ such blocks. Adjacent iff $x_1 = x_2$, thus only the diagonal blocks are non-zero. Since the multiplicity is the multiplicity in Y, diagonal blocks are just B.

Definition 5.10. Define operator $C : L^2(X ⓩ Y) \to L^2(X)$ by

$$Cf(x) = \sum_{y \in V_Y} f(x, y)$$

In other words, C sums over clouds.

Remark
Note $X ⓩ Y$, Z, and H all have the same vertex set, thus $L^2(X ⓩ Y), L^2(Z), L^2(H)$ are identical.

Definition 5.11. $\tilde{\beta} \in L^2(X ⓩ Y)$ is constant on clouds if $\tilde{\beta}(x, y_1) = \tilde{\beta}(x, y_2)$ for all $x \in V_X$, $y_1, y_2 \in V_Y$.

Proposition 5.12. If $\tilde{\beta} \in L^2(X ⓩ Y)$ is constant on clouds, then $AC\tilde{\beta} = d_X C\tilde{A}\tilde{\beta}$

Proof

- Recall $Af(v) = \sum_{w \in V_X} A_{v,w} f(w)$
- Let $v \in V_X$, $\tilde{e_v}$ be the vector in $L^2(X ⓩ Y)$ defined by $\tilde{e_v} = 1$ if $v = x$ and $\tilde{e_v} = 0$ if $v \neq x$.
- Since $\tilde{e_v}$ form a basis for the subspace of vectors of $L^2(X ⓩ Y)$ that are constant on clouds, suffice to show $AC\tilde{e_v} = d_X C\tilde{A}\tilde{e_v}$ for all $v \in V_X$
- Note both sides are in $L^2(X)$. Let $w \in V_X$.

- $C\tilde{e_v}() = \sum_{y\in V_Y} \tilde{e_v}(\ ,y) = |Y|\delta_v() = d_X\delta_v()$ (here $\delta()$ is a function that gives 1 if input is v and 0 otherwise, second equality by definition of $\tilde{e_v}$)
- $AC\tilde{e_v}(w) = d_X A\delta_v(w) = d_X \sum_{t\in V_x} A_{t,w}\delta_v(t) = d_X\cdot$ #edges between v and w
- $$\begin{aligned}(C\tilde{A}\tilde{e_v})(w) &= \sum_{y\in V_Y} (\tilde{A}\tilde{e_v}(w,y) \\ &= \sum_{y\in V_Y}\sum_{(p,q)\in H} \tilde{A}_{(w,y)(p,q)}\tilde{e_v}(p,q) \\ &= \sum_{y\in V_Y}\sum_{q\in V_Y} \tilde{A}_{(w,y)(v,q)}\tilde{e_v}(v,q) \\ &\text{note adjacency in } H \text{ requires } L_v(q) = L_w(y) = \text{some edge between } v \text{ and } w \\ &\text{since } L_v, L_w \text{ are bijective, there is only one pair of } q,y \text{ and one edge, for fixed } v,w. \\ &\text{Hence, the adjacency is reduced to adjacency of } w,v \\ &\text{Also, summing over } Y \text{ ensures all adjacency is captured} \\ &= \#\text{ edges between } w,v\end{aligned}$$

5.3 Eigenvalues of Zig-Zag Product

This section gives an upper bound on the $\lambda(X \text{Ⓩ} Y)$ in terms of $\lambda(X), \lambda(Y)$.

Definition 5.13.

- For any vertex $v \in V_X$, define $\tilde{f}_v \in L^2(X \text{Ⓩ} Y)$ by
$$\tilde{f}_v(x,y) = \begin{cases} |Y|^{-1/2} & v = x \\ 0 & v \neq x \end{cases}$$
So $\tilde{f}_v$ is a unit vector that is constant on the v-cloud and 0 outside the v-cloud, and is orthogonal to each other
- For any $\tilde{\alpha} \in L_0^2(X \text{Ⓩ} Y)$ and any $v \in V_X$, define
  - $\tilde{\alpha}_v^{\|} = \langle\tilde{\alpha}, \tilde{f}_v\rangle\tilde{f}_v$
  - $\tilde{\alpha}^{\|} = \sum_{v\in V_X} \tilde{\alpha}_v^{\|}$ (constant on clouds part)
  - $\tilde{\alpha}^{\perp} = \tilde{\alpha} - \tilde{\alpha}^{\perp}$ (summing to 0 on clouds part)
- 0 denotes the zero vector in $L_0^2(X \text{Ⓩ} Y)$

Remark

- By definition, $\sum_{x,y\in X\times Y}\tilde{\alpha}(x,y)=0$ (summing to 0)
- $\tilde{\alpha}^{\|}(x,y)=\langle\tilde{\alpha},\tilde{f}_x\rangle\tilde{f}_x(x,y)=\langle\tilde{\alpha},\tilde{f}_x\rangle|Y|^{-1/2}$, the part constant on clouds.
- $\langle\tilde{\alpha},\tilde{f}_x\rangle=\sum_{(p,q)\in X\times Y}\tilde{\alpha}(p,q)\overline{\tilde{f}(p,q)}=\sum_{q\in Y}\sum_{p\in X}\tilde{\alpha}(p,q)\tilde{f}_x(p,q)=\sum_{q\in Y}\tilde{\alpha}(x,q)|Y|^{-1/2}$
- With fixed x,

$$\begin{aligned}\sum_{y\in Y}\tilde{\alpha}^{\perp}(x,y)&=\sum_{y\in Y}\tilde{\alpha}(x,y)-\tilde{\alpha}^{\|}(x,y)\\&=\sum_{y\in Y}\tilde{\alpha}(x,y)-\langle\tilde{\alpha},\tilde{f}_x\rangle|Y|^{-1/2}\\&=\left(\sum_{y\in Y}\tilde{\alpha}(x,y)\right)-|Y|^{1/2}\langle\tilde{\alpha},\tilde{f}_x\rangle\\&=\left(\sum_{y\in Y}\tilde{\alpha}(x,y)\right)-\sum_{q\in Y}\tilde{\alpha}(x,q)\\&=0\end{aligned}$$

  the part sums to 0 on clouds

- Also, $\langle\tilde{\alpha}^{\|},\tilde{\alpha}^{\perp}\rangle=0$

$$\begin{aligned}\langle\tilde{\alpha}^{\|},\tilde{\alpha}^{\perp}\rangle&=\langle\tilde{\alpha}^{\|},\tilde{\alpha}\rangle-\langle\tilde{\alpha}^{\|},\tilde{\alpha}^{\|}\rangle\\&=\sum_{v\in X}\langle\tilde{\alpha},\tilde{f}_v\rangle\langle\tilde{f}_v,\tilde{\alpha}\rangle-\|\tilde{\alpha}^{\|}\|^2\\&=\sum_{v\in X}\|\tilde{\alpha}_v^{\|}\|^2-\|\tilde{\alpha}^{\|}\|^2=0\end{aligned}$$

Lemma 5.14. Let $\tilde{\alpha}\in L^2(X \textcircled{z} Y)$. Then $C\tilde{\alpha}^{\perp}=0$

Proof
Actually this is already shown in the previous remark using $\langle\tilde{\alpha},\tilde{f}_x\rangle$. Here is another proof using $\langle\tilde{\alpha}^{\perp},\tilde{f}_x\rangle$
For any $v\in V_X$, first note

$$\begin{aligned}\langle\tilde{\alpha}^{\perp},\tilde{f}_v\rangle&=\sum_{x,y\in X\times Y}\tilde{\alpha}^{\perp}(x,y)\overline{\tilde{f}_v(x,y)}\\&=\sum_{y\in Y}\tilde{\alpha}^{\perp}(v,y)|Y|^{-1/2}\\&=|Y|^{-1/2}\sum_{y\in Y}\tilde{\alpha}^{\perp}(v,y)\end{aligned}$$

Then

$$\begin{aligned}(C\tilde{\alpha}^{\perp})(v) &= \sum_{y\in Y}\tilde{\alpha}^{\perp}(v,y)\\ &= |Y|^{1/2}\langle\tilde{\alpha}^{\perp},\tilde{f}_v\rangle\end{aligned}$$

Also note

$$\begin{aligned}\langle\tilde{\alpha}^{\perp},\tilde{f}_v\rangle &= \langle\tilde{\alpha}-\sum_{x\in X}\langle\tilde{\alpha},\tilde{f}_x\rangle\tilde{f}_x,\tilde{f}_v\rangle\\ &= \langle\tilde{\alpha},\tilde{f}_v\rangle-\sum_{x\in X}\langle\tilde{\alpha},\tilde{f}_x\rangle\langle\tilde{f}_x,\tilde{f}_v\rangle\\ &= \langle\tilde{\alpha},\tilde{f}_v\rangle-\langle\tilde{\alpha},\tilde{f}_v\rangle = 0\end{aligned}$$

Since $\langle\tilde{f}_x,\tilde{f}_v\rangle = 0$ iff $x \neq v$ and $\|\tilde{f}_v\| = 1$

Lemma 5.15. If $\tilde{\beta} \in L^2(X ⓩ Y)$, then $\|\tilde{A}\tilde{\beta}\| = \|\tilde{\beta}\|$.

Proof
$\tilde{A}\tilde{\beta}(x,y) = \sum_{p,q\in X\times Y}\tilde{A}_{(x,y)(p,q)}\tilde{\beta}(p,q) = \tilde{\beta}(p,q)$, since H is 1-regular. Thus $\tilde{A}$ is essentially a permutation. Then, as $\|\|$ in our case sums over all vertices, there is no difference.

Lemma 5.16. For any $\tilde{\alpha} \in L^2_0(X ⓩ Y)$, we have

$$|\langle\tilde{M}\tilde{\alpha},\tilde{\alpha}\rangle| \leq d_Y^2|\langle\tilde{A}\tilde{\alpha}^{\|},\tilde{\alpha}^{\|}\rangle| + 2d_Y\|\tilde{\alpha}^{\|}\|\cdot\|\tilde{B}\tilde{\alpha}^{\perp}\| + \|\tilde{B}\tilde{\alpha}^{\perp}\|^2$$

Proof
First note, if $\tilde{\beta}$ is constant on clouds, then $\tilde{B}\tilde{\beta} = d_Y\tilde{\beta}$

$$\begin{aligned}\tilde{B}\tilde{\beta}(x,y) &= \sum_{p,q\in X\times Y}\tilde{B}_{(p,q)(x,y)}\tilde{\beta}(p,q)\\ &= \sum_{q\in Y}\tilde{B}_{(x,q)(x,y)}\tilde{\beta}(x,q)\\ &\text{since adjacency in } H \text{ requires } x = p\\ &= d_Y\tilde{\beta}(x,y)\end{aligned}$$

since $\tilde{\beta}$ is constant on clouds

Then

$$
\begin{aligned}
\langle \tilde{M}\tilde{\alpha}, \tilde{\alpha}\rangle &= \langle \tilde{B}\tilde{A}\tilde{B}\tilde{\alpha}, \tilde{\alpha}\rangle \\
&\quad \text{by Corollary 5.8} \\
&= \langle \tilde{A}\tilde{B}\tilde{\alpha}, \tilde{B}\tilde{\alpha}\rangle \\
&\quad \text{since } \tilde{B} \text{ is real symmetric hence self-adjoint} \\
&= \langle \tilde{A}\tilde{B}(\tilde{\alpha}^{\|} + \tilde{\alpha}^{\perp}), \tilde{B}(\tilde{\alpha}^{\|} + \tilde{\alpha}^{\perp})\rangle \\
&= \langle \tilde{A}\tilde{B}\tilde{\alpha}^{\|}, \tilde{B}\tilde{\alpha}^{\|}\rangle + \langle \tilde{A}\tilde{B}\tilde{\alpha}^{\|}, \tilde{B}\tilde{\alpha}^{\perp}\rangle + \langle \tilde{A}\tilde{B}\tilde{\alpha}^{\perp}, \tilde{B}\tilde{\alpha}^{\|}\rangle + \langle \tilde{A}\tilde{B}\tilde{\alpha}^{\perp}, \tilde{B}\tilde{\alpha}^{\perp}\rangle \\
&\quad \text{by observation, and since } \tilde{\alpha}^{\|} \text{ constant on clouds} \\
&= d_Y^2\langle \tilde{A}\tilde{\alpha}^{\|}, \tilde{\alpha}^{\|}\rangle + d_Y\langle \tilde{A}\tilde{\alpha}^{\|}, \tilde{B}\tilde{\alpha}^{\perp}\rangle + d_Y\langle \tilde{A}\tilde{B}\tilde{\alpha}^{\perp}, \tilde{\alpha}^{\|}\rangle + \langle \tilde{A}\tilde{B}\tilde{\alpha}^{\perp}, \tilde{B}\tilde{\alpha}^{\perp}\rangle
\end{aligned}
$$

Then, by triangle inequality

$$
\begin{aligned}
|\langle \tilde{M}\tilde{\alpha}, \tilde{\alpha}\rangle| &\leq d_Y^2|\langle \tilde{A}\tilde{\alpha}^{\|}, \tilde{\alpha}^{\|}\rangle| + d_Y\|\tilde{A}\tilde{\alpha}^{\|}\| \cdot \|\tilde{B}\tilde{\alpha}^{\perp}\| + d_Y\|\tilde{A}\tilde{B}\tilde{\alpha}^{\perp}\| \cdot \|\tilde{\alpha}^{\|}\| + \|\tilde{A}\tilde{B}\tilde{\alpha}^{\perp}\| \cdot \|\tilde{B}\tilde{\alpha}^{\perp}\| \\
&\quad \text{by Lemma 5.15, } \|\tilde{A}\tilde{\beta}\| = \|\tilde{\beta}\| \\
&= d_Y^2|\langle \tilde{A}\tilde{\alpha}^{\|}, \tilde{\alpha}^{\|}\rangle| + 2d_Y\|\tilde{\alpha}^{\|}\| \cdot \|\tilde{B}\tilde{\alpha}^{\perp}\| + \|\tilde{B}\tilde{\alpha}^{\perp}\|^2
\end{aligned}
$$

Lemma 5.17. If Y is nonbipartite, then for any $\tilde{\alpha} \in L_0^2(X \circledZ Y)$, we have

$$\|\tilde{B}\tilde{\alpha}^{\perp}\| \leq \lambda(Y)\|\tilde{\alpha}^{\perp}\|$$

Proof

- For any $v \in X$, define $\alpha_v^{\perp} \in L^2(Y)$ by $\alpha_v^{\perp}(y) = \tilde{\alpha}^{\perp}(v, y)$
- Then, following convention in Def 1.10, we can write $\tilde{\alpha}^{\perp} = (\alpha_{v_1}^{\perp} \mid \alpha_{v_2}^{\perp} \mid \dots \mid \alpha_{v_n}^{\perp})^T$, where $n = |X|$. That is,
$$\tilde{\alpha}^{\perp}(x, y) = \alpha_{v_1}^{\perp}(y)\delta_{v_1}(x) + \alpha_{v_2}^{\perp}(y)\delta_{v_2}(x) + \cdots + \alpha_{v_n}^{\perp}(y)\delta_{v_n}(x)$$
- Note $\sum_{y\in Y} \tilde{\alpha}_v^{\perp}(y) = \sum_{y\in Y} \tilde{\alpha}^{\perp}(v, y) = 0$ since $\tilde{\alpha}^{\perp}$ sums up to 0 on clouds, then $\alpha_v^{\perp} \in L_0^2(Y)$
- $\tilde{B}\tilde{\alpha}^{\perp} = (B\alpha_{v_1}^{\perp} \mid B\alpha_{v_2}^{\perp} \mid \dots \mid B\alpha_{v_n}^{\perp})^T$, since
$$
\begin{aligned}
\tilde{B}\tilde{\alpha}^{\perp}(x, y) &= \sum_{p,q\in X\times Y} \tilde{B}_{(x,y)(p,q)}\tilde{\alpha}^{\perp}(p, q) \\
&= \sum_{q\in Y} \tilde{B}_{(x,y)(x,q)}\tilde{\alpha}^{\perp}(x, q) \\
&= \sum_{q\in Y} B_{y,q}\tilde{\alpha}_x^{\perp}(q) \\
&\quad \text{essentially the x-block of } \tilde{B}, \text{ which is } B \\
&= B\tilde{\alpha}_x^{\perp}(y)
\end{aligned}
$$

That is,

$$\tilde{B}\tilde{\alpha}^{\perp}(x,y) = \tilde{B}\tilde{\alpha}^{\perp}_{v_1}(y)\delta_{v_1}(x) + \tilde{B}\tilde{\alpha}^{\perp}_{v_2}(y)\delta_{v_2}(x) + \cdots + \tilde{B}\tilde{\alpha}^{\perp}_{v_n}(y)\delta_{v_n}(x)$$

- With similar arguments as Prop 1.26 and Prop 3.6, since Y is nonbipartite and $\alpha^{\perp}_v \in L^2_0(Y)$, we can show that $\|B\alpha^{\perp}_{v_j}\| \leq \lambda(Y)\|\alpha^{\perp}_{v_j}\|$

- Hence

$$\begin{aligned}\|\tilde{B}\tilde{\alpha}^{\perp}\|^2 &= \|B\alpha^{\perp}_{v_1}\|^2 + \|B\alpha^{\perp}_{v_2}\|^2 + \cdots + \|B\alpha^{\perp}_{v_n}\|^2 \\ &\leq \lambda(Y)^2 \left(\|\alpha^{\perp}_{v_1}\|^2 + \|\alpha^{\perp}_{v_2}\|^2 + \cdots + \|\alpha^{\perp}_{v_n}\|^2\right) \\ &= \lambda(Y)^2\|\tilde{\alpha}^{\perp}\|^2\end{aligned}$$

since both $B\alpha^{\perp}_v$ and $\alpha^{\perp}_v$ are orthgonal "basis"

Lemma 5.18. $\tilde{\beta}_1, \tilde{\beta}_2 \in L^2(X \textcircled{z} Y)$ s.t. $\tilde{\beta}_2$ is constant on clouds. Then $\langle C\tilde{\beta}_1, C\tilde{\beta}_2\rangle = d_X\langle\tilde{\beta}_1, \tilde{\beta}_2\rangle$

Proof

$$\begin{aligned}\langle C\tilde{\beta}_1, C\tilde{\beta}_2\rangle &= \sum_{x\in X}\left(\sum_{y\in Y}\tilde{\beta}_1(x,y)\right)\left(\overline{\sum_{y\in Y}\tilde{\beta}_2(x,y)}\right) \\ &= |Y|\sum_{x\in X}\left(\sum_{y\in Y}\tilde{\beta}_1(x,y)\overline{\tilde{\beta}_2(x,y)}\right) \\ &= d_X\sum_{x,y\in X\times Y}\tilde{\beta}_1(x,y)\overline{\tilde{\beta}_2(x,y)} \\ &= d_X\langle\tilde{\beta}_1, \tilde{\beta}_2\rangle\end{aligned}$$

Lemma 5.19. If X is nonbipartite, the for any $\tilde{\alpha} \in L^2_0(X \textcircled{z} Y)$, we have

$$|\langle\tilde{A}\tilde{\alpha}^{\parallel}, \tilde{\alpha}^{\parallel}\rangle| \leq \frac{\lambda(X)}{d_X}\langle\tilde{\alpha}^{\parallel}, \tilde{\alpha}^{\parallel}\rangle$$

Proof

First note that $C\tilde{\alpha} = C\tilde{\alpha}^{\|}$, since $C\tilde{\alpha}^{\perp} = 0$. Then,

$$\begin{aligned}
\langle \tilde{A}\tilde{\alpha}^{\|}, \tilde{\alpha}^{\|}\rangle &= \frac{\langle C\tilde{A}\tilde{\alpha}^{\|}, C\tilde{\alpha}^{\|}\rangle}{d_X} \\
&\text{Lemma 5.18, } \tilde{\alpha}^{\|} \text{ constant on clouds} \\
&= \frac{\langle AC\tilde{\alpha}^{\|}, C\tilde{\alpha}^{\|}\rangle}{d_X^2} \\
&\text{Prop 5.12, } \tilde{\alpha}^{\|} \text{ constant on clouds} \\
&= \frac{\langle AC\tilde{\alpha}, C\tilde{\alpha}\rangle}{d_X^2} \\
&\text{Note, or essentially because } \tilde{\alpha}^{\|} \text{ constant on clouds}
\end{aligned}$$

Since $\tilde{\alpha} \in L_0^2(X Ⓩ Y)$, $C\tilde{\alpha} \in L_0^2(X)$, then by Prop 3.6 (a version of Rayleigh-Ritz),

$$|\langle \tilde{A}\tilde{\alpha}^{\|}, \tilde{\alpha}^{\|}\rangle| = \frac{\langle AC\tilde{\alpha}, C\tilde{\alpha}\rangle}{d_X^2} \leq \frac{\lambda(X)\langle C\tilde{\alpha}, C\tilde{\alpha}\rangle}{d_X^2} = \frac{\lambda(X)\langle C\tilde{\alpha}^{\|}, C\tilde{\alpha}^{\|}\rangle}{d_X^2} = \frac{\lambda(X)\langle \tilde{\alpha}^{\|}, \tilde{\alpha}^{\|}\rangle}{d_X}$$

Theorem 5.20. Nonbipartite $|Y|$-regular graph X, nonbipartite d_Y-regular graph Y, then for any choice of labeling from X to Y, we have

$$\lambda(X Ⓩ Y) \leq \frac{d_Y^2 \lambda(X)}{d_X} + d_Y \lambda(Y) + \lambda(Y)^2$$

Proof
Let $\tilde{\alpha} \in L_0^2(X Ⓩ Y)$, then

$$\begin{array}{rcccccc}
|\langle \tilde{M}\tilde{\alpha}, \tilde{\alpha}\rangle| & \leq & d_Y^2 |\langle \tilde{A}\tilde{\alpha}^{\|}, \tilde{\alpha}^{\|}\rangle| & + & 2d_Y \|\tilde{\alpha}^{\|}\| \cdot \|\tilde{B}\tilde{\alpha}^{\perp}\| & + & \|\tilde{B}\tilde{\alpha}^{\perp}\|^2 \\
& \leq & \frac{d_Y^2}{d_X}\lambda(X)\|\tilde{\alpha}^{\|}\|^2 & + & 2d_Y\lambda(Y)\|\tilde{\alpha}^{\|}\| \cdot \|\tilde{\alpha}^{\perp}\| & + & \lambda(Y)^2\|\tilde{\alpha}^{\perp}\|^2 \\
& & \text{Lemma 5.19} & & \text{Lemma 5.17} & & \text{Lemma 5.17}
\end{array}$$

Let $p = \frac{\|\tilde{\alpha}^{\|}\|}{\|\tilde{\alpha}\|}$, $q = \frac{\|\tilde{\alpha}^{\perp}\|}{\|\tilde{\alpha}\|}$. Since $\langle \tilde{\alpha}^{\|}, \tilde{\alpha}^{\perp}\rangle = 0$, we have $p^2 + q^2 = 1$, thus $p \leq 1, q \leq 1$. Also, $0 \leq (p-q)^2 = 1 - 2pq$, so $2pq \leq 1$. Therefore,

$$\frac{|\langle \tilde{M}\tilde{\alpha}, \tilde{\alpha}\rangle|}{\langle \tilde{\alpha}, \tilde{\alpha}\rangle} \leq d_Y^2 \lambda(X) p^2 + 2\lambda(Y) d_Y pq + \lambda(Y)^2 q^2 \leq \frac{d_Y^2 \lambda(X)}{d_X} + d_Y \lambda(Y) + \lambda(Y)^2$$

Since $\tilde{\alpha}$ is arbitrary, by Rayleigh-Ritz, we have

$$\lambda(X Ⓩ Y) \leq \frac{d_Y^2 \lambda(X)}{d_X} + d_Y \lambda(Y) + \lambda(Y)^2$$

5.4 An Actual Expander Family

Definition 5.21. Base Graph

- Let p be a prime number greater than 35, $G = \mathbb{Z}_p^8$
- Define identity of G as 0
- Define $\gamma : \mathbb{Z}_p^2 \to G$ by $\gamma(x,y) = (x, xy, xy^2, xy^3, xy^4, xy^5, xy^6, xy^7)$ Let $\Gamma = \{\gamma(x,y) \,|\, (x,y) \in \mathbb{Z}_p^2\}$
- Note $\Gamma \overset{s}{\subseteq} G$. Let $W = \mathrm{Cay}(G, \Gamma)$
- Let $\omega = e^{\frac{2\pi i}{p}}$

Remark
For inverse of $\gamma(x,y)$, consider $\gamma(x^{-1}, y^{-1})$. Note in $\mathbb{Z}_p$, we have $y^{-1}x^{-1} = p - y + p - x = x^{-1}y^{-1}$

Definition 5.22. Write an element $a \in G = \mathbb{Z}_p^8$ as $a = (a_0, a_1, \ldots, a_7)$. For two elements $a, b \in G$, define $a \cdot b = a_0b_0 + a_1b_1 + \cdots + a_7b_7$

Lemma 5.23.

$$\sum_{x \in G} \omega^{c \cdot x} = \begin{cases} p^8 & c = 0 \\ 0 & c \neq 0 \end{cases}$$

Proof
Since $|G| = p^8$, the result is immediate if $c = 0$. If $c \neq 0$, then $c_j \neq 0$ for some j. WLOG, assume $c_0 \neq 0$, then

$$\begin{aligned} \sum_{x \in G} \omega^{c \cdot x} &= \sum_{x \in G} \omega^{c_0x_0}\omega^{c_1x_1} \ldots \omega^{c_7x_7} \\ &= \sum_{x_0 \in \mathbb{Z}_p} \sum_{x_1 \in \mathbb{Z}_p} \cdots \sum_{x_7 \in \mathbb{Z}_p} \omega^{c_0x_0}\omega^{c_1x_1} \ldots \omega^{c_7x_7} \\ &= \left(\sum_{x_0 \in \mathbb{Z}_p} \omega^{c_0x_0} \right) \left(\sum_{x_1 \in \mathbb{Z}_p} \cdots \sum_{x_7 \in \mathbb{Z}_p} \omega^{c_1x_1} \ldots \omega^{c_7x_7} \right) = 0 \end{aligned}$$

By Lemma 1.18,

$$\sum_{j=0}^{p-1} (\omega^{c_0})^j = 0$$

since $c_0 \neq 0$ and $c_0 \in \mathbb{Z}_p$, thus p not dividing c_0.

Definition 5.24.

- For any $a \in G$, define $f_a \in L^2(W)$ by $f_a(x) = \omega^{a \cdot x}$

- Let A be the adjacency operator of W, define $\lambda_a = \sum_{x,y\in\mathbb{Z}_p} \omega^{a\cdot\gamma(x,y)}$

Lemma 5.25. For all $a \in G$, $Af_a = \lambda_a f_a$

Remark
That is, eigenvalues of W are precisely the numbers λ_a

Proof
Recall formula for adjacency operator in Cayley graph:

$$(Af)(g) = \sum_{\gamma\in\Gamma} f(g\gamma)$$

For any $b \in G$, we have

$$\begin{aligned}(Af_a)(b) &= \sum_{\gamma(x,y)\in\Gamma} f_a(b+\gamma(x,y)) \\ &= \sum_{x,y\in\mathbb{Z}_p} \omega^{a\cdot b + a\cdot\gamma(x,y)} \\ &= \omega^{a\cdot b} \sum_{x,y\in\mathbb{Z}_p} \omega^{a\cdot\gamma(x,y)} \\ &= \lambda_a f_a(b)\end{aligned}$$

Lemma 5.26. $\{f_a \,|\, a \in G\}$ is an orthogonal set.

Remark
That is, every f_a is an orthogonal eigenvector of W with eigenvalue λ_a. Also, the number of f_a is exactly $|W|$, thus forming an orthogonal eigenbasis.

Proof
For $a \neq b$

$$\begin{aligned}\langle f_a, f_b\rangle &= \sum_{x\in G} f_a(x)\overline{f_b(x)} \\ &= \sum_{x\in G} \omega^{a\cdot x}\omega^{-b\cdot x} \\ &= \sum_{x\in G} \omega^{(a-b)\cdot x} = 0\end{aligned}$$

Since $a - b \neq 0$ and Lemma 5.23

Lemma 5.27. For all $a \neq 0$, we have $0 \leq \lambda_a \leq 7p$

Proof
Fact: a nonzero polynomial of degree d with coefficients in $\mathbb{Z}_p$ has no more than d roots in $\mathbb{Z}_p$

$$\begin{aligned}\lambda_a &= \sum_{x,y\in\mathbb{Z}_p} \omega^{a\cdot\gamma(x,y)} \\ &= \sum_{x,y\in\mathbb{Z}_p} \omega^{a_0x+a_1xy+\cdots+a_7xy^7} \\ &= \sum_{y\in\mathbb{Z}_p}\sum_{x\in\mathbb{Z}_p} \omega^{(a_0+a_1y+\cdots+a_7y^7)x}\end{aligned}$$

Define $g(t) = a_0 + a_1t + \cdots + a_7t^7$ as a polynomial in $\mathbb{Z}_p$. For any fixed $y \in \mathbb{Z}_p$, by Lemma 1.18, we have

$$\sum_{x=0}^{p-1}(\omega^{g(y)})^x = \begin{cases} p & g(y) = 0 \\ p & g(y) \neq 0 \end{cases}$$

since $|Z_p| = p$ and p divides $g(y)$ iff $g(y) = 0$ in Z_p
Hence $\lambda_a = np \geq 0$, where n is the number of roots of g in $\mathbb{Z}_p$. Since $a \neq 0$ and $g(t)$ has 7 coefficients, by fact and by $n \leq 7$, we have $\lambda_a \leq 7p$.

Lemma 5.28. Let $W = \mathrm{Cay}(G, \Gamma)$ in Def. 5.21, then

- $|W| = d_W^4$
- W is nonbipartite
- $\lambda(W) < \frac{d_W}{5} = \frac{p^2}{5}$

Remark
A graph in the Expander family need not be nonbipartite, but we do need nonbipartite here to control the eigenvalue of zig-zag product.

Proof

- $|W| = |G| = p^8$, $d_W = |\Gamma| = p^2$
- By Lemma 5.27 and Lemma 5.25, all eigenvalues of W is positive, so $-p^2$ is not an eigenvalue. By Prop. 1.16, not bipartite.
- By Lemma 5.27 and the fact that $p > 35$

$$\lambda(W) \leq 7p \leq \frac{p^2}{5}$$

Definition 5.29. Let $W = \mathrm{Cay}(G, \Gamma)$ in Def. 5.21, recursively define a sequence $\{W_n\}$ as follows

- $W_1 = W^2$
- $W_{n+1} = W_n^2 ⓩ W$

where $W_n^2 ⓩ W$ is formed using any choice of labeling from W to W_n^2

Remark
We have $d_{W_1} = d_W^2 = (p^2)^2 = p^4$, so $d_{W_1^2} = (p^4)^2 = p^8 = |W|$, the zig-zag product is well-defined for first step. Also, $d_{W_2} = (d_W)^2 = p^4$, so inductively $d_{W_n} = p^4$ and all zig-zag products are well-defined. This legitimacy uses 8 in $G = \mathbb{Z}_p^8$.

Theorem 5.30. Let $\{W_n\}$ be as in Def. 5.29, then $\{W_n\}$ is an expander family.

Proof

- Since $|W_{n+1}| = |W_n^2 \times W| = |W_n|^2|W|$ and $|W| = p^8$, we have $|W_n| \to \infty$
- WTS: $\lambda(W_n) < 2p^4/5$ for all n, then the spectral gap of W_n is at least $3p^4/5$ for all n, so $\{W_n\}$ is an expander family.
- Note W_n^2 is nonbipartite, since square of any undirected graph with at least one edge will contain a loop. If there is an edge between a, b in X, then there will be a loop at a
- By Prop. 1.33, Lemma 5.28

  $$\lambda(W^2) < \frac{p^4}{25} \leq \frac{2p^4}{5}$$

  Base case done. For inductive step, assume $\lambda(W_n) < 2p^4/5$. By Lemma 5.29, we know W is nonbipartite. Then,

  $$\begin{aligned}\lambda(W_n^2 ⓩ W) &\leq \frac{p^4\lambda(W_n^2)}{p^8} + p^2\lambda(W) + \lambda(W)^2 \\ &\text{by Theorem 5.20, and by nonbipartite } W_n^2, W \\ &\leq \frac{4p^4}{25} + p^2 \cdot \frac{p^2}{5} + \frac{p^4}{25} \\ &\text{by Lemma 5.29 and inductive hypothesis} \\ &= \frac{2p^4}{5}\end{aligned}$$

5.5 Zig-Zag Product and Semidirect Product

Definition 5.31. Let G, K be finite groups.

- Let $\theta : K \to \mathrm{Aut}(G)$ be a homomorphism
- Let $\gamma \in G$ s.t. $\gamma^2 = e_G = 1$. That is, γ is an element of order 2 in G, and following convention of semidirect product we take $e_G = e_K = 1$, implicitly assuming G, K are subgroups of a larger group.
- Let $\Gamma = \{\theta_k(\gamma) \mid k \in K\}$. Note each element of Γ is its own inverse; hence, $\Gamma \Subset G$.
- Let $\Lambda \Subset K$.
- We have Cayley graphs $X = \mathrm{Cay}(G, \Gamma)$ and $Y = \mathrm{Cay}(K, \Lambda)$
- For each $g \in G$ (also each vertex in $\mathrm{Cay}(G, \Gamma)$), let E_g be as in Def. 5.2
- Define $L_g : K \to E_g$ by taking $L_g(k)$ to be the edge between g and $g\theta_k(g)$, let L be the labeling defined by these L_g

Proposition 5.32. Following Def. 5.31,

$$\mathrm{Cay}(G, \Gamma) \textcircled{z}_L \mathrm{Cay}(H, \Lambda) = \mathrm{Cay}(G \rtimes_\theta H, \Upsilon),$$

where Υ is the multiset $\{\sigma_1\gamma\sigma_2 \mid (\sigma_1, \sigma_2) \in \Lambda \times \Lambda\}$

Proof

Let Z, H be as in Section 2. First claim: zig-zag graph Z equals $\mathrm{Cay}(G \rtimes K, \Lambda)$. Same vertex set is $G \times K$. Consider the multiplicity of the edge between $(g_1, k_1), (g_2, k_2)$.

- In Z, we have $g_1 = g_2$, and the multiplicity is the edge between k_1, k_2, which is the multiplicity of σ in Λ s.t. $k_1\sigma = k_2$, essentially the multiplicity of $k_1^{-1}k_2$ in Λ
- In $\mathrm{Cay}(G \rtimes K, \Lambda)$, we have $(g_1, k_1)(1, \sigma) = (g_2, k_2)$, thus $g_1 = g_2$ and $\sigma = k_1^{-1}k_2$. Multiplicity essentially lies on the multiplicity of $k_1^{-1}k_2$ in Λ

Second claim: hyphen graph H equals $\mathrm{Cay}(G \rtimes K, \{\gamma\})$. The Cayley graph is well-defined since $\gamma^2 = 1$ thus $\{\gamma\} \Subset G \rtimes K$. Same vertex set is $G \times K$. Note both graphs are 1-regular.

- $(g_1, k_1)(g_2, k_2)$ are adjacent in H
- iff $L_{g_1}(k_1) = L_{g_2}(k_2)$
- iff $\{$edge between g_1 and $g_1\theta_{k_1}(\gamma)\}$ same as $\{$edge between g_2 and $g_2\theta_{k_2}(\gamma)\}$

- iff $g_1 = g_2\theta_{k_2}(\gamma)$ and $g_2 = g_1\theta_{k_1}(\gamma)$ (since if $g_1 = g_2$, we have $k_1 = k_2$ by bijectivity of L_x)
- iff $g_2 = g_1\theta_{k_1}(\gamma)$ and $\theta_{k_1}(\gamma)\theta_{k_2}(\gamma) = 1$
  - $\theta_{k_1}(\gamma) = \theta_{k_2}(\gamma)$
- iff $g_2 = g_1\theta_{k_1}(\gamma)$ and $k_1 = k_2$ (since each element of Γ is its own inverse, last line suggests the two elements are essentially the same)
- iff $(g_1, k_1)(\gamma, 1) = (g_2, k_2)$ in $G \rtimes K$
- iff (g_1, k_1) and (g_2, k_2) are adjacent in $\mathrm{Cay}(G \rtimes K, \{\gamma\})$

Since $\mathrm{Cay}(G,\Gamma)ⓩ_L\mathrm{Cay}(H,\Lambda) = Z \cdot H \cdot Z$ (Prop. 5.6) and $\mathrm{Cay}(G \rtimes_\theta H, \Upsilon) = \mathrm{Cay}(G \rtimes K, \Lambda) \cdot \mathrm{Cay}(G \rtimes K, \{\gamma\}) \cdot \mathrm{Cay}(G \rtimes K, \Lambda)$ (Prop. 1.35), done.

Example
Let Y be d_Y-regular with vertices $y_1, y_2, \ldots, y_n$, X be the graph with single vertex K and n loops. Then, $XⓩY = Y^2$, with any labeling from Y to X.

Remark
Can relate to Def 5.29, where W_1 defined as W^2 is essentially a zig-zag product.

Proof
Can use $Y = K^2$ as an illustrating example for proof ideas.

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